

BCSE406 – NOSQL DATABASES

Project Ideas

1. **Polyglot persistence demo app** — an e-commerce site using Redis for sessions/cart (key-value), MongoDB for the product catalog (document), and Neo4j for "customers who bought this also bought..." (graph). Great for showing *why* you'd pick different databases for different jobs, not just *how*.
2. **CAP theorem visualizer** — spin up a small Cassandra or Riak cluster, deliberately induce network partitions (tc/iptables), and build a dashboard showing consistency vs. availability trade-offs in real time as you tune replication settings.
3. **Fraud-ring detector** — a graph database project (Neo4j) that models transactions as a graph and surfaces suspicious cyclical/clustered patterns — a classic, resume-friendly graph use case.
4. **Relational-to-NoSQL schema migration tool** — takes a normalized SQL schema + sample queries and proposes a denormalized document or column-family schema, explaining the trade-offs it made.
5. **IoT time-series store** — column-family design (Cassandra/HBase) with TTL-based data expiration and wide-row modeling for sensor data, benchmarked against a naive relational approach.
6. **Key design linter** — a CLI tool that scans key-value access logs and flags hot-spotting risks, inconsistent delimiters, or overly long keys (ties directly into your Chapter 3 material).
7. **Cross-database benchmark suite** — same workload (e.g., a social feed) run against key-value, document, column-family, and graph databases, with a report on latency, throughput, and where each one breaks down.
8. Graph-Based Financial Fraud Detection using Neo4j
9. Smart Hospital Information System (MongoDB + Neo4j)
10. Distributed Cyber Threat Intelligence Platform
11. AI-Powered Smart Traffic Management System
12. Supply Chain Traceability using Multi-Model NoSQL
13. Industrial IoT Predictive Maintenance Platform

14. Personalized Learning Analytics System
15. Blockchain-Backed Immutable Audit Database
16. Adaptive Multi-Model NoSQL Database for Smart Cities
17. AI-Driven Autonomous NoSQL Query Optimization Framework

Title

1. Hybrid Multi-Model NoSQL Database for Smart Hospitals
2. Adaptive Auto-Sharding Engine for IoT Data
3. AI-Based Dynamic Schema Evolution Framework
4. Graph-Based Financial Fraud Detection
5. Intelligent Disaster Response Data Platform
6. Smart City Digital Twin using Multi-Model NoSQL
7. Privacy-Preserving Federated NoSQL Learning Platform
8. Blockchain-Backed Immutable NoSQL Audit System
9. Self-Optimizing Multi-Cloud NoSQL Storage Manager
10. AI-Driven Autonomous NoSQL Query Optimizer